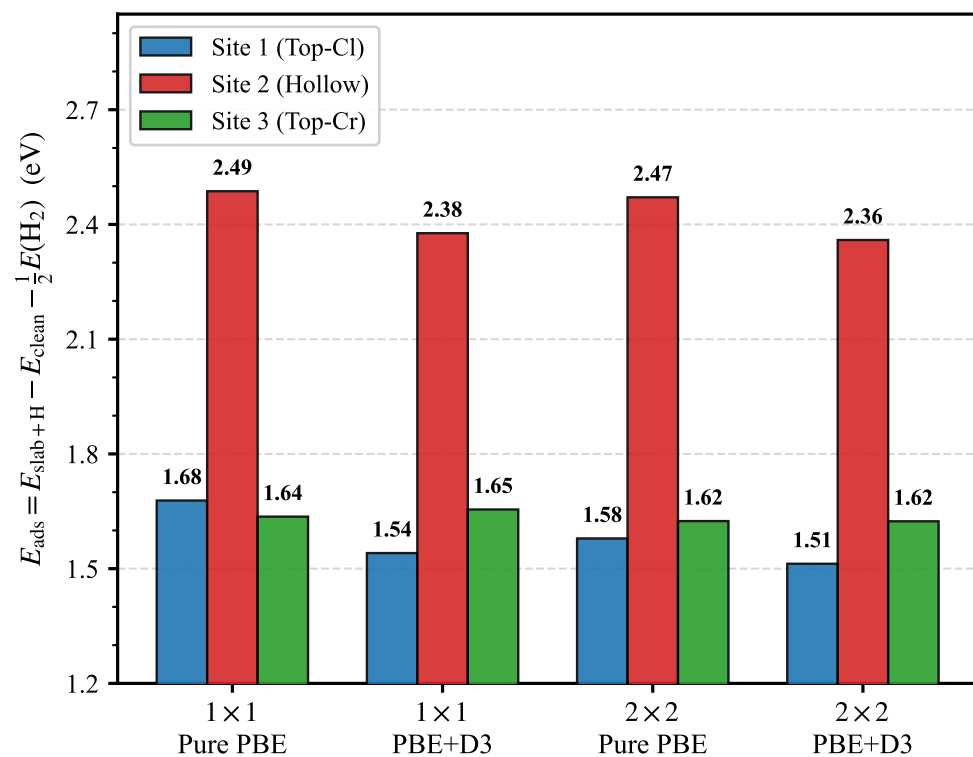
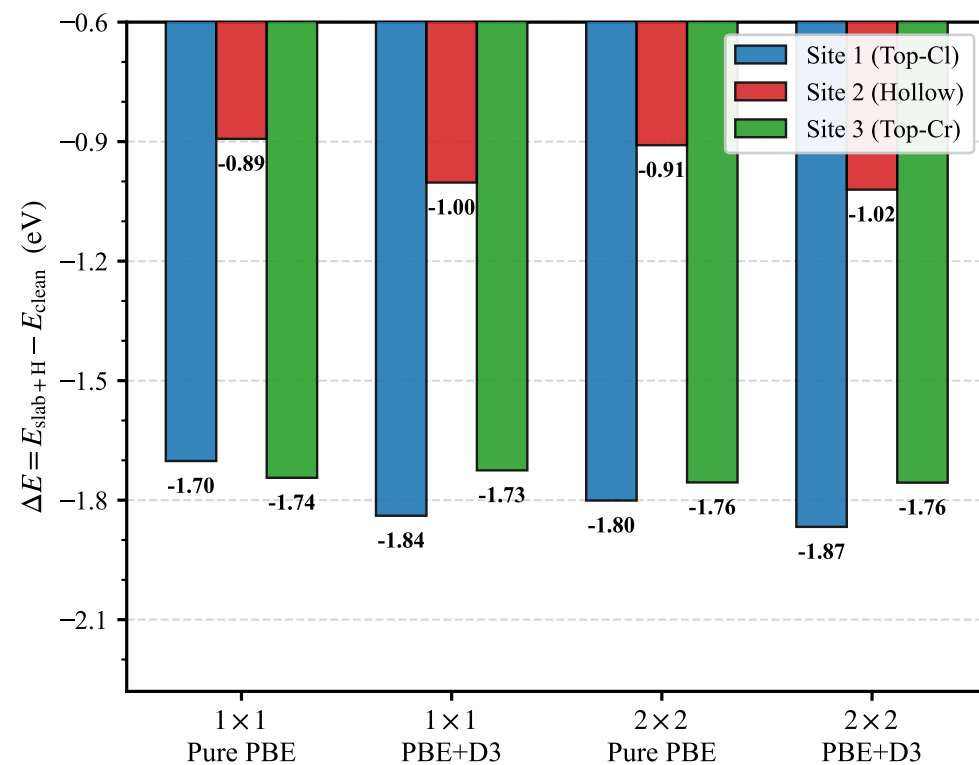


Hydrogen Adsorption Thermodynamics on Monolayer CrCl₃: Multi-Scale & Dispersion Benchmarks

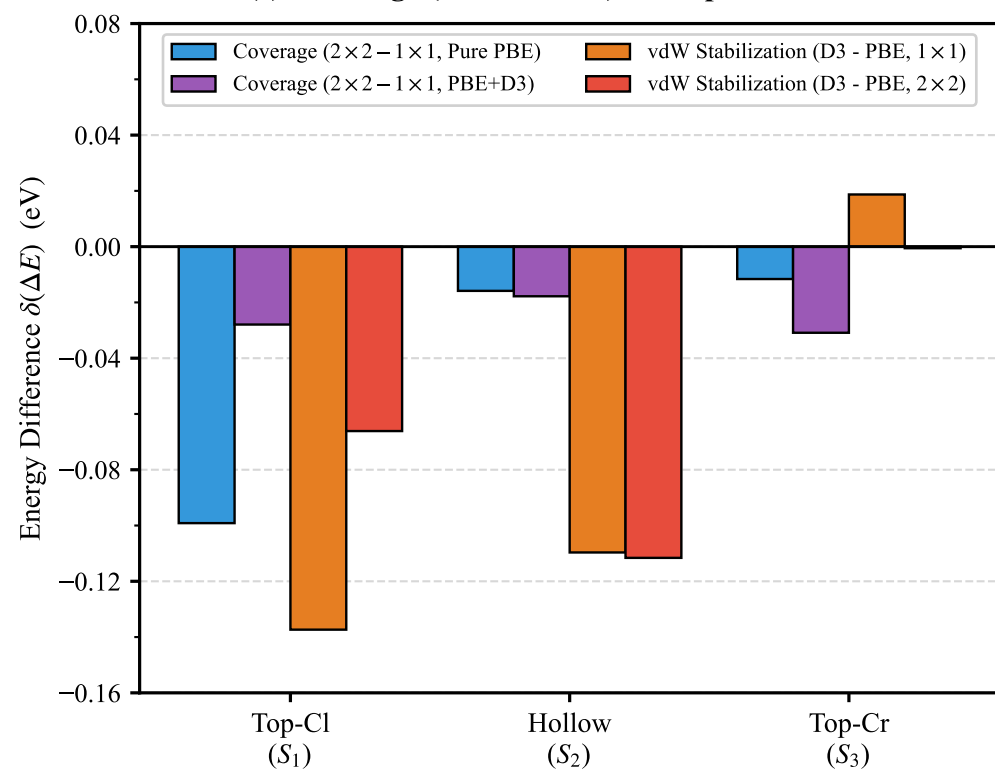
(a) Hydrogen Adsorption Energy vs. Isolated $\frac{1}{2}\text{H}_2$



(b) Thermodynamic Binding Energy ΔE



(c) Coverage (2×2 vs 1×1) & Dispersion Shifts



(d) HER Free Energy Profile ($\Delta G_{\text{H}^*} = E_{\text{ads}} + 0.24$ eV)

